

BRANDVLEI, South Africa

WMO: 685230

Lat: 30.4647S

Lon: 20.4783E

Elev: 922

StdP: 90.73

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.9	0.6	-15.7	1.1	12.8	-13.8	1.3	11.4	13.2	1.4	9.7	17.7	1.2	200	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.8	38.0	17.2	36.5	16.8	35.2	16.3	20.1	28.5	19.4	28.4	18.7	28.9	4.7	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	14.3	23.5	16.6	13.2	23.8	15.0	11.9	23.7	62.1	28.1	59.5	27.9	56.8	28.7	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.5	7.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.1	26.9	26.2	24.3	19.9	15.1	11.1	10.8	12.3	16.0	19.9	22.5	24.9
	DBStd	6.78	3.17	3.20	3.40	3.94	3.77	3.16	3.74	3.81	3.89	4.51	4.17	3.30
	HDD10.0	81	0	0	0	0	4	22	35	17	2	0	0	0
	HDD18.3	931	0	1	3	31	112	217	233	191	92	37	13	1
	CDD10.0	3410	523	454	442	297	163	56	61	89	182	306	375	462
	CDD18.3	1218	264	222	187	78	13	1	1	5	21	85	138	204
	CDH23.3	14474	3314	2539	2010	733	187	16	22	103	324	1040	1662	2524
CDH26.7	7229	1860	1352	976	258	39	1	0	23	107	478	799	1336	
Wind	WSAvg	3.5	4.2	3.8	3.3	3.0	2.8	3.0	3.2	3.2	3.4	3.9	4.2	4.3
Precipitation	PrecAvg	137	14	17	26	21	8	7	7	3	4	13	11	7
	PrecMax	377	76	92	116	136	59	36	75	29	33	85	72	52
	PrecMin	25	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	85	21	24	28	27	11	9	14	6	8	20	17	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	38.9	37.2	33.7	29.9	25.5	25.6	29.5	32.4	36.7	37.5	38.9
		MCWB	17.9	18.3	17.1	16.5	14.3	12.3	11.8	12.8	14.1	16.4	15.9	16.7
	2%	DB	38.2	37.0	35.4	31.4	27.5	22.7	23.5	26.3	29.6	34.0	35.1	36.8
		MCWB	17.3	17.8	16.8	15.6	13.4	10.9	10.9	11.7	13.1	15.2	15.0	16.5
	5%	DB	36.5	35.3	33.8	29.6	25.3	20.6	21.4	23.6	27.0	31.9	33.2	35.2
		MCWB	17.0	17.0	16.3	15.0	12.6	10.0	9.9	10.4	11.9	14.3	14.7	16.1
	10%	DB	34.7	33.7	32.0	27.5	23.0	18.0	19.0	20.8	24.3	29.3	31.3	33.4
		MCWB	16.4	16.4	15.8	14.3	11.9	9.4	9.1	9.7	10.6	13.3	14.0	15.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	21.2	20.4	18.3	16.1	13.1	12.6	14.6	15.7	17.7	18.2	19.8
		MCDB	29.3	30.9	27.5	26.9	24.9	21.9	23.1	21.7	30.4	32.4	30.9	30.9
	2%	WB	20.0	20.3	19.1	17.0	14.8	11.9	11.6	12.9	13.5	16.6	17.0	18.4
		MCDB	27.8	28.9	28.4	27.1	22.6	19.9	21.7	22.4	26.8	29.9	29.2	30.1
	5%	WB	19.2	19.6	18.3	16.3	13.9	11.0	10.6	11.5	12.4	15.4	15.7	17.5
		MCDB	29.1	27.8	28.3	26.3	22.9	18.0	19.8	22.1	25.0	28.5	29.6	30.1
	10%	WB	18.3	18.9	17.4	15.5	12.8	10.1	9.5	10.0	11.3	14.1	14.7	16.7
		MCDB	30.5	27.6	28.4	25.2	22.0	16.5	17.5	19.6	22.9	27.0	28.6	29.9
Mean Daily Temperature Range	5% DB	MDBR	14.8	14.2	14.6	13.5	14.2	13.7	14.8	15.9	15.2	15.3	15.2	15.7
		MCDBR	16.5	15.7	16.0	15.5	17.0	17.4	17.9	19.4	18.7	18.1	17.3	17.4
		MCWBR	5.4	5.4	5.7	6.0	7.6	8.6	9.0	9.2	8.7	6.8	6.6	6.1
	5% WB	MCDBR	12.8	11.8	12.7	13.1	14.3	14.0	15.5	16.4	16.9	16.6	15.8	14.9
		MCWBR	4.4	4.8	5.2	5.3	6.6	7.5	8.0	7.9	8.1	6.5	6.9	5.8
Clear-Sky Solar Irradiance	taub	0.309	0.319	0.311	0.292	0.273	0.260	0.263	0.292	0.307	0.318	0.295	0.299	
	taud	2.558	2.536	2.552	2.603	2.642	2.641	2.616	2.491	2.415	2.420	2.556	2.570	
	Ebn at Noon	1032	1003	979	950	926	916	927	934	966	992	1039	1045	
	Edh at Noon	108	107	99	85	73	68	73	93	111	119	108	108	
All-Sky Solar Radiation	RadAvg	8.53	7.70	6.52	5.03	3.93	3.40	3.72	4.65	6.07	7.37	8.40	8.92	
	RadStd	0.29	0.30	0.21	0.21	0.18	0.15	0.14	0.19	0.22	0.22	0.30	0.36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.04	N/A	N/A	N/A	N/A

Nomenclature: See separate page